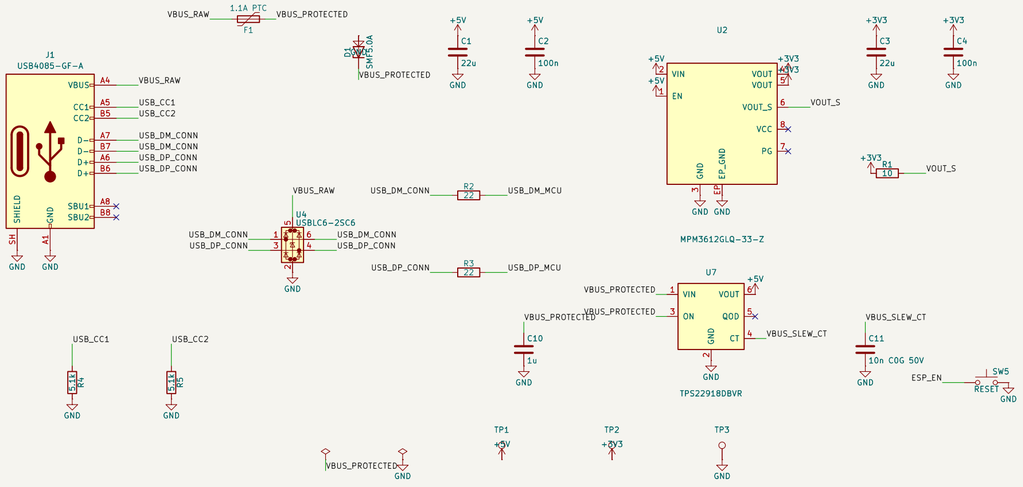
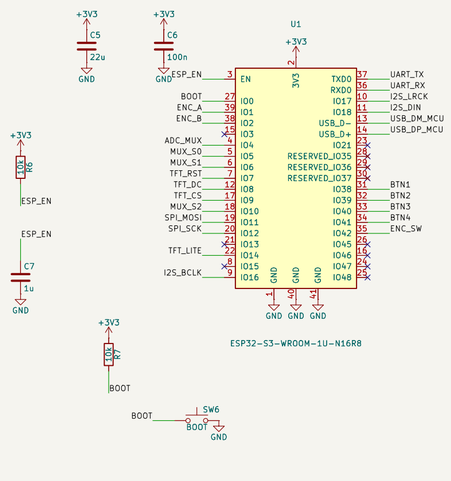
USB-C / input protection

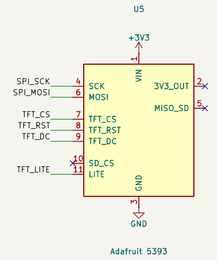


3.3V integrated power module

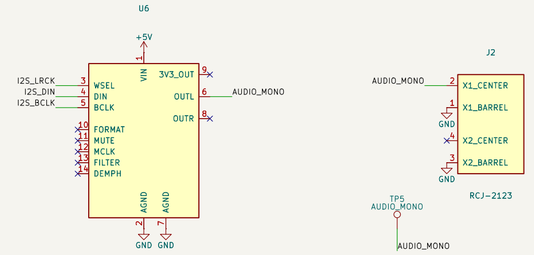
ESP32-S3-WROOM-1U N16R8



Display 5393 – 320 x 172

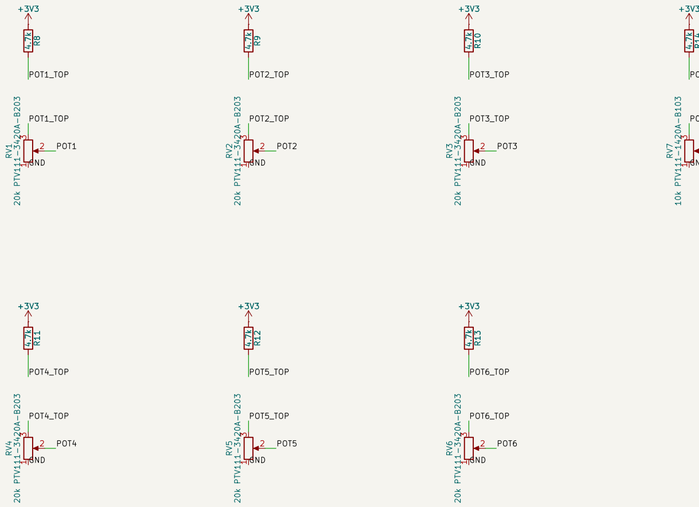


PCM5102 line output – mono

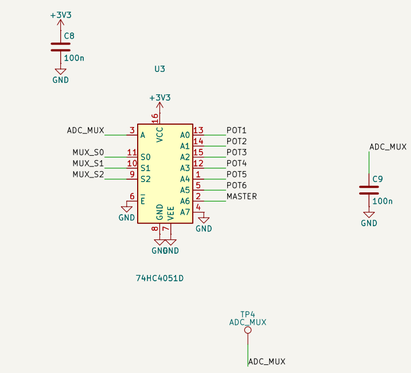


X1 = mono out. X2 center unused. Both barrels grounded.

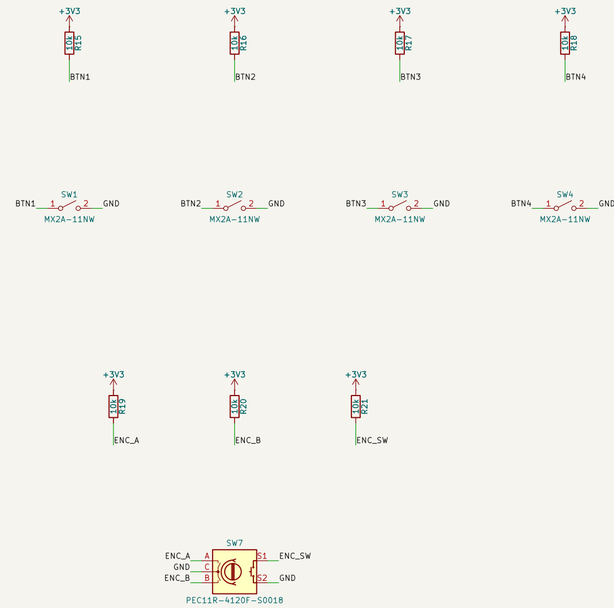
Seven analog controls



One 8:1 analog multiplexer



Performance switches / encoder



PROTOTYPE – all controls populated; USB power policy and bench qualification pending

MPM: no external inductor / MOSFET / feedback divider. PG and optional VCC capacitor unused.

Display VIN = 3.3V because TFTCS / RESET pull-ups connect to VIN. SD interface unused.

ADC: 4.7k per pot high end. 100nF at ADC1 GPIO4. Initial prototype scan dwell 7ms per channel.

Audio: one DAC output only; no L/R summing. Minimum external load 1k.

Sheet: /		
File: ESP-DubSiren-V2.kicad_sch		
Title:		
Size: A2	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/1